



Recommendations with Benefits: Exploring Explanations in Information Sharing Recommender Systems for Temporary Teams

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ABSTRACT

Increased use of collaborative technologies and agile teamwork models has led to a greater need for temporary teams. Unfortunately, they lack the normal team formation processes that traditional teams use. Information sharing recommender systems can be used to share information about team members amongst the team; however, these systems rely on the team members themselves to disclose valuable information. While prior research has shown that an effective way to encourage user disclosure is through explanations to the user about what benefits they will gain from disclosure, the timing of such explanations has yet to be considered. In a between-subjects study with 150 participants, we assessed the content and timing of explanations on levels of disclosure in temporary teams. Our results indicate that providing benefit-related explanations during the time of disclosure can increase user disclosure, and providing benefit-related explanations during the recommendation process can increase user trust in the system. These results provide important design implications for teams and the HCI community.

KEYWORDS

Recommender systems;
information sharing; mental
models; explanations

1. Introduction

Most recently, the concept of temporary teams has grown in popularity due to the ability of organizations to shuffle and deliberately compose a team to accomplish a set of specific tasks (Dalal et al., 2017; Tannenbaum et al., 2012). These kinds of teams help organizations remain agile, with the ability to add and remove members based on the team's needs (Tannenbaum et al., 2012). Unfortunately, the greatest strength of these teams, which is their agile temporality and rapid task completion (Bouas & Arrow, 1996), is also their greatest weakness when concerning team mental models (TMMs), as these teams are not afforded the time commonly needed to establish the norms, processes, and expectations that form TMMs (McNeese & Reddy, 2017). Team formation is often a complex and difficult task that organizations repeatedly undergo, and these formations are generally not rapid or smooth as teams need time to learn about each other and establish common understandings to complete work (Tuckman & Jensen, 1977). Specifically, this time and effort contribute to teams' and teammates' abilities to form TMMs, which help teammates understand and navigate their relationships with their teammates and predict their actions and interactions (Converse et al., 1993). Given their importance to team performance (Cannon-Bowers & Salas, 1998), teammates often spend a great deal of time intentionally forming and familiarizing themselves with each other to form these models as they often foster and grow through repeated interaction (Mohammed et al., 2010). In

turn, temporary teams are either constantly in flux or in existence for too short of a time to improve based on past performance or interactions with teammates.

To address this issue, recent work has identified the benefit of information sharing for these teams (Webber et al., 2019), which work has theorized could be accelerated through information sharing technologies (Freeman & Wohn, 2019). Most notably, information sharing recommender systems, which parse through user profiles and share information amongst users (Liang et al., 2009), are poised to aid temporary teams in their ability to form TMMs rapidly (Musick et al., 2023). However, the implementation of these systems is still early, and research has yet to assess the potential design challenges that often face recommender systems that could impact this novel teaming environment. Most notably, privacy concerns, which commonly impact the amount of information humans share with recommender systems (Jozani et al., 2020), stand to directly impact the effectiveness of these information sharing recommender systems in temporary teams. Specifically, the direct function of these systems is to share information from one teammate to another, which can trigger data-sharing privacy concerns for the users (Greif & Sarin, 1987). These privacy concerns that restrict sharing stand to blacken the effectiveness of these systems directly and these concerns may already be heightened due to the unfamiliar and temporary nature of these teams.

To mitigate these privacy concerns, this article details a study that explores the design of explanations of

information sharing recommender system recommendations, specifically the content and timing of these explanations, as these designs have been shown to directly impact the role of privacy in other recommender system implementations (Tintarev & Masthoff, 2011; Vig et al., 2009). Particularly, the content of explanations has been manipulated to either detail the benefits or rationale behind information sharing, both of which can impact the information humans choose to disclose (Herlocker et al., 2000; Symeonidis et al., 2009). For timing, this study either presented explanations during disclosure, which could impact disclosure habits, or during recommendation receipt, which could impact perceived privacy (Tintarev & Masthoff, 2011). Holistically, this study will explore the impact of explanations on disclosure and the perception of privacy in the use of information sharing recommender systems, which will inform the design of explanations leveraged by information sharing recommender systems. Most importantly, these results can facilitate the rapid sharing of team-relevant information across teammates without being hindered by unexpected privacy concerns.

In sum, temporary teams stand to greatly benefit from technologies that can catalyze the formation of TMMs, and information sharing recommender systems show great promise in being this catalyst. Unfortunately, privacy concerns stand to hinder the effectiveness of these systems. In response, this study utilized a between-subjects online vignette study ($n=150$) to explore the design of explanations to benefit information sharing recommender systems in temporary teams, which is done by answering the following research questions:

RQ1: Does the timing of a recommendation explanation (i.e., during disclosure vs. recommendation receipt) affect the user's disclosure and perceptions of an information-sharing recommender system?

RQ2: Does the content of the explanation (i.e., benefit vs. algorithmic) affect the user's disclosure and perceptions of an information-sharing recommender system?

This study makes several contributions to recommender systems, teamwork, and HCI research. First, a better understanding of how to promote personal information disclosure between teammates is important to supporting temporary teams as this information sharing is associated with increased team performance (Druskat & Kayes, 2000). Second, this study provides insights into how the content and occurrence of explanations for information sharing recommender systems impact users' self-disclosure of the system and affect their perceptions of that system in regard to trust and privacy concerns. Third, design implications presented in this study support HCI researchers interested in developing an information sharing recommender system that not only encourages users to disclose valuable, team-relevant information but also promotes trustworthy perceptions of said systems. The article will next review the relevant literature that lays the groundwork for this study, followed by the methodology of the experiment and data analysis. Finally, it will discuss how the results answer the research questions and the resulting design implications.

2. Background

This section will provide an overview of the current literature and research gaps that motivate the study including (1) the concept of TMMs and HCI research in regard to temporary teams; (2) the use of recommender systems in improving teamwork; and (3) the role of explanations in the context of recommender systems.

2.1. Team formation and team mental models

An important factor in a team's ability to adapt in a coordinated manner is its effectiveness in building and maintaining compatible mental models that result in shablack expectations for the team (Converse et al., 1993). Mental models permit individuals to pblackict future events, and *teammate* TMMs (e.g., preferences, skills, tendencies) are used to pblackict the actions and needs of one's teammates (Rouse & Morris, 1986). This ability for team members to pblackict each other's actions is important because it can lead to a significant increase in team performance (Klimoski & Mohammed, 1994; Mohammed et al., 2010). These models, however, are difficult to construct in temporary teams for a multitude of reasons. For one, these teams are composed of unknown teammates without past experiences to draw upon (Lv & Feng, 2021). They then have little time to develop these shablack experiences and models, because the teams generally work together for a short period of time (Son & Rojas, 2011) and disband once their task is complete (Bakker et al., 2013). As a result, HCI and Computer Supported Cooperative Work (CSCW) research has attempted to develop tools and strategies to overcome these challenges (Harris et al., 2019).

Some recent HCI and CSCW research has focused on ways to accelerate and support the development of TMMs in temporary teams. For instance, some research has investigated how to selectively choose team members in order to increase familiarity (Salehi et al., 2017), balance personalities (Lykourantzou et al., 2016), or increase diversity (Gómez-Zará et al., 2020). However, unfamiliar teams often do not have the privilege of being picky about their team members (Brouthers et al., 1995) which has motivated research to support temporary teams *after* team formation. This body of research includes leveraging tools, such as team training, which has been shown to increase team effectiveness (Rice et al., 2007). Meanwhile, other research has shown that when temporary teams utilize reflection tools there is improved shablack understanding and *teammate* TMMs (Mulder et al., 2004). Similarly, other researchers have emphasized the importance of designing team-building games that can promote team cohesion (Tuffley, 2008).

Along with these studies, adjacent research has suggested using technology to share teammate information (Freeman & Wohn, 2019; Webber et al., 2019). Research has shown that when teammates share personal information amongst the team, there is a significant positive impact on the team's performance (Druskat & Kayes, 2000). Initial research in using technology to share such information has shown promise. In a study involving fast-response spontaneous

virtual teams, researchers found that sharing teammate skill information between team members on temporary teams had a significant positive impact on perceived task cohesion and team performance (Yang et al., 2015). Recommender systems have been proven to be capable of predicting and influencing user behaviors (Zengy et al., 2022) and thus are a promising possibility for team information sharing. Importantly, other research has highlighted that, although temporary team members perceive the sharing of such information as helpful, there is a need to alleviate the privacy concerns of users (Musick et al., 2023). Therefore, this study aims to fill this research gap by better understanding how explanations can be used to promote positive system perceptions.

2.2. Recommender systems in teamwork

While information sharing recommender systems are relatively novel to teams, recommender systems are not an unfamiliar technology to teams. Generally speaking, recommender systems are an important technology for suggesting items or information to users (Panagiotakis et al., 2021; Resnick & Varian, 1997; Ricci et al., 2015). A specifically helpful and complex type of recommender system important to this study is the group recommender system. A group recommender system makes recommendations to a defined group of users (Jameson & Smyth, 2007). The groups that these systems focus on can fall into multiple categories. The two categories relevant to unfamiliar teams are occasional groups and automatically identified groups. Occasional groups perform an occasional activity together with a common goal (Garcia et al., 2011), while automatically identified groups are selected based on preferences and/or available resources (Twyman et al., 2022). In both of these instances, the group members might know little to nothing about their team members and not spend enough time working together to naturally learn about one another. Furthermore, the transient nature of these teams may exacerbate a recommender system's performance heterogeneity, where a subset of users may receive much more accurate recommendations based on uneven user history (Zhang et al., 2023).

In using group recommender systems, each individual user has to consider the benefits they receive from self-disclosure to the system (Jozani et al., 2020). Users consider the benefits they will gain as a trade-off to giving up a certain level of privacy (Xu et al., 2013). Recommender systems themselves can be designed in such a way that nudges users toward more self-disclosure (Knijnenburg, 2015). This design concept can alleviate the burden of the user in calculating the benefits and risks of self-disclosure to the system (Benisch et al., 2011). In order for this to work, the system design needs to consider a variety of factors associated with an individual's decision to disclose. First and foremost, individual differences play a large role in one's decision to self-disclose (Knijnenburg et al., 2013). Users also heavily consider the context within which the information is being shared (Bansal et al., 2016) and the type of information being shared (Mehdy et al., 2021). The latter is of

particular concern to temporary teams and TMMs, as research shows that individuals will limit sharing certain information that may give teammates a negative impression of themselves (Brennan et al., 2009). This way of using a recommender system places users simultaneously as producers and consumers of the recommendations (Burke et al., 2022) and posits unique challenges for encouraging disclosure to the system.

The body of HCI literature points toward intelligent recommender systems as a potential solution. Though a system could be designed as a blunt instrument to share any and all teammate information which results in maximum disclosure, these types of systems are likely to create large privacy concerns (Musick et al., 2023). Recommender systems that are trusted by users need to consider and incorporate the values of those users, including privacy (Stray et al., 2022). Research in group recommender systems has found benefits in allowing users to selectively disclose different types of information (Najafian et al., 2020) as simply providing users with privacy controls can decrease privacy concerns (Tsai et al., 2010). However, these control settings inevitably result in users disclosing less information to the system and, as a result, the group (Najafian et al., 2020). Therefore, design aspects of the system (e.g., explanations) must be considered in finding ways to promote disclosure while limiting privacy concerns.

2.3. The role and design of explanations

As noted above, recommender systems can be designed in ways that encourage a greater level of self-disclosure from users, and these designs will be especially important for teams where multiple individuals with different privacy concerns have to make individual decisions about disclosure. Particularly, an essential element of recommender systems is the explanations they provide about the recommendations they make. When recommender systems provide explanations as to why a recommendation was made, users better understand and are more likely to accept the recommendation (Tran et al., 2019; Wilkinson et al., 2021; Zhang & Chen, 2020). It is important to note that explanations are not simply justifications of a choice as explanations are typically designed to fulfill multiple purposes (Nothdurft et al., 2013). Other purposes include satisfaction, transparency, scrutability, trust, persuasiveness, effectiveness, and efficacy (Kouki et al., 2019; Lu et al., 2023; Sardianos et al., 2021; Tintarev, 2007; Tintarev & Masthoff, 2009). A great deal of research has focused on how to achieve these aims, but recommender systems designed to benefit a group, as opposed to an individual, possess several additional challenges in deciding when and what to explain (Najafian & Tintarev, 2018).

Recommendation explanations can provide transparency to the user about several aspects of the recommendation and are categorized by the nature of the explanation produced (Himeur et al., 2021). First, explanations in this context must consider what content the explanation should contain. In some recommender systems, explanations focus

on describing *how* a recommendation was generated (Vig et al., 2009), which has been shown to substantially increase user trust (Herlocker et al., 2000). However, these explanations which are often algorithmic in nature might have trade-offs. For instance, some research has highlighted that an algorithmic explanation that aims to increase transparency might also decrease trust depending on the user and the context (Knijnenburg & Kobsa, 2013; Tintarev & Masthoff, 2012). Meanwhile, other recommender systems have emphasized describing *why* a recommendation was generated through justifications (Symeonidis et al., 2009). In the context of group recommender systems, a justification explanation might be able to clarify why a recommendation was provided to the group and the benefit the recommendation has for the group (Najafian et al., 2020).

Recommender system research has shown the value of combining more than one explanation type in tailoring recommendations for specific users (Sardianos et al., 2020), and thus a second type of explanation for group contexts should consider when the explanation should occur. The process of generating recommendations for the purpose of positively influencing teamwork possesses two occurrences for when explanations may be provided: (1) in eliciting information from team members and (2) in providing recommendations to the team members. Most group recommender systems are designed to provide explanations *during* the user's receipt of the recommendation (Tintarev & Masthoff, 2011; Tran et al., 2019). This is similar to how it is done for single-user recommender systems, as explanations provided with recommendations can result in increased trust, perceived effectiveness, and satisfaction (Tintarev & Masthoff, 2011). However, recent research has highlighted the importance of users understanding the disclosure benefit early on in the process to increase disclosure (Najafian et al., 2023). Previous work has shown that recommender systems can effectively use the timing of explanations to increase the amount of information users disclose (Knijnenburg & Kobsa, 2013) and their acceptance of those recommendations (Varlamis et al., 2023). Importantly, more research is necessary to understand the trade-offs between when these explanations should occur and what information those explanations should provide.

3. Methodology

3.1. Experimental design

The research questions for this study focus on how the occurrence and content of an explanation might influence user disclosure to an information sharing recommender system and user perceptions of the system. Given that this work is explicitly interesting in the timing and content of explanations, we have elected to use a theoretical recommender system presented via a vignette study, meaning recommendations and corresponding explanations were generated and held consistent across participants. To thoroughly explore this and to ensure that the occurrence and content of the explanations do not confound the investigation, this study utilizes a $2 \times 2 \times 2$ (all between-subjects)

design that simulates for the participants the receipt of recommendations for working with new teammates on a temporary team based on information they would have disclosed to the system.

3.1.1. Content manipulation

The first manipulation is a manipulation of content. The content of explanations for previous recommender systems typically involves algorithmic explanations (e.g., how was this recommendation produced) (Tintarev, 2007). However, group recommender system research emphasizes that perceived disclosure benefit can increase disclosure (Najafian et al., 2023), a tactic that has been demonstrated in consumer research that takes into account the consumer's perceived utility of a product in making recommendations (Ghanem et al., 2022). In the information age, users constantly choose how much information to disclose based on what they think they are gaining from accessing the computer system and/or social platform (Himeur et al., 2021). Thus, the content for an explanation could involve describing disclosure benefits. As indicated in the literature review, a decent amount of research into how to get users to accept recommendations has been done, but benefit-related explanations are uniquely poised to encourage users to self-disclose to a system and to assess the extent to which this is true, we chose to contrast it against a control condition. This results in two content conditions, including (a) algorithmic rationale or (b) benefit rationale. In other words, participants either receive a justification of what variables the system considered in making its recommendation or an explanation of how the recommender system's recommendations will benefit the user and the team. These justifications were manually generated by researchers beforehand, and they were consistently presented via vignettes in an online survey.

3.1.2. Timing manipulations

This study leverages two separate manipulations to explore timing; as such, the second and third manipulations relate to the occurrence of the explanation. Prior research typically describes explanations occurring *after* or *during* the recommendation (Tintarev & Masthoff, 2011; Tran et al., 2019). However, if the goal is to increase the amount disclosed and decrease privacy concerns during the disclosure process, it might be beneficial to provide an explanation during disclosure selection, before the system provides any recommendations. To explore this, both occurrence variables were manipulated as binary conditions (i.e., yes or no) regarding whether or not an explanation occurred at that time. A summary of the manipulations is shown in Table 1. This design not only allows us to compare the timing of the explanations as during disclosure during recommendation but also, in one condition, as a repeated variable. Similar to explanation content, all participants placed within the same timing condition received the same explanations, and a real recommender system did not generate these explanations.

3.2. Procedure

In this section, we will provide an overview of the online vignette study procedure used to conduct this online factorial survey study, including pre-surveys, vignettes, disclosure selection, and measurement of perceptions of the system.

3.2.1. Pre-surveys

The task began with participants completing informed consent and pre-surveys. Part of the pre-surveys involved an abbreviated Big Five survey that measured their personality on eight of the 30 Big Five sub-facets. These eight facets were selected based on the results of a preliminary study by the researchers, in which all sub-facets were rated for

perceived helpfulness by 155 survey participants. To elicit more diverse preferences, responses, and perceptions from participants, some facets were chosen due to the expectation that users will perceive them as more or less helpful and more or less sensitive based on these preliminary results:

3.2.2. Video vignette

Participants then watched and read from a video vignette (always the same for each condition) that described a worker who had been assigned to a 6-week temporary team in the workplace where none of the team members knew each other (see Figure 1 for screenshot). This vignette was created by the second author and was chosen to incorporate a multimodal comprehension strategy of the vignette, which is more effective for inserting someone inside a story (Lenters & Winters, 2013). In addition to describing the team's overall tasks and the worker's role in helping the team accomplish them, the video also described the technology being used (i.e., a technology designed and developed to intelligently share information about team members to each other to accelerate their understanding of one another to improve team efficiency and performance). A knowledge check was also embedded in the video for future use during the survey in combination with additional attention checks.

Table 1. Study 4 2 × 2 × 2 experimental design.

Manipulation 1: Explanation content (between)			
1 Algorithmic rationale			
2 Benefit rationale			
Manipulation 2: When?—During disclosure (between)			
1 No			
2 Yes			
Manipulation 3: When?—During recommendation (between)			
1 No			
2 Yes			
Study 4: Design matrix			
Conditions	Manipulations		
	Content	During disclosure	During Recs
1	Algorithmic	Yes	Yes
2	Algorithmic	Yes	No
3	Algorithmic	No	Yes
4	Algorithmic	No	No
5	Benefit	Yes	Yes
6	Benefit	Yes	No
7	Benefit	No	Yes
8	Benefit	No	No

Conditions 4 and 8 were combined to make a baseline condition as neither has explanations provided at either occurrence.

3.2.3. Disclosure selection

Next, participants were prompted to place themselves in this situation and to make selections regarding what information they would be willing to disclose to the system to inform recommendations to their teammates. This is the point in the experiment where participants actively make decisions about how much personal information they want to disclose to the recommender system. These selections were based on



Figure 1. Vignette screen shot.

their own results for the eight personality facets, see Table 2 (presented in a randomized order).

3.2.4. Explanations and recommendations

After participants made their selections, they were presented with four recommendations for working better with their hypothetical teammates (presented in a randomized order). Attributes used for recommendations included dependability, assertiveness, self-efficacy, and trust (see Table 3 for an example of what recommendations might look like). For some conditions, explanations were provided at this point, although the contents of the explanation varied by condition. An example for each content type based on when the explanation occurs is provided in Table 4. After completing this portion of the task, participants then completed post-task measures related to their perceptions of the system.

3.3. Participants and demographics

Before participant recruitment and experimentation, the author team obtained institutional review board approval. Participants for this study were recruited using Prolific, an online platform designed for recruiting participants for online research studies (Palan & Schitter, 2018). A power analysis was performed to determine the number of participants required for a $2 \times 2 \times 2$ ANOVA involving main and interaction effects. This analysis determined that 146 participants would be required to reach a reasonable power for a medium effect size. A total of 150 participants were recruited, which resulted in 18–19 participants per

condition. We applied the following criteria in recruiting participants: must be a resident of the United States, English must be their first language and must work as part of a group at work [based on established recommendations (Aguinis et al., 2021)]. Attention checks were included within the vignette study with the purpose of filtering out participants who failed more than one check (Berinsky et al., 2014); however, no participants failed more than one attention check and had to be removed. The average age of participants was 35.09 ($SD = 10.61$). Seventy-six participants identified as women, 69 identified as men, four identified as non-binary or third gender, 1 preferred not to disclose, and 0 preferred not to specify. Regarding team experience, 73.3% of participants indicated that they had experience working on ongoing project teams, 37.3% indicated that they had experience working on temporary project teams, and 20.0% indicated that they had experience working on temporary action teams. Participants were paid \$3.50 as an incentive for completing the survey of approximate length 20 min (\$10.50 as their hourly rate, which is above the minimum incentive recommended (Brawley & Pury, 2016)).

3.4. Measurements

The measures used in this study include disclosure selections and system perception measures to answer both research questions. Descriptions of each of the measures used are provided in the following sections.

3.4.1. Demographics and personal measures

Participant demographics were measured for descriptive purposes. These measures included their age, education, race, language, gender, employment, and team experience. Additionally, *Trust Propensity* was collected, which measures how likely an individual is to trust a person or thing using four Likert-scale items (Jessup et al., 2019).

Table 2. Preliminary results for perceptions based on the selected personality type sub-facets.

Helpfulness	Sharing	Personality facet 1	Personality facet 2
More helpful	Share more	Dependability	Assertiveness
Less helpful	Share more	Self-efficacy	Trust
More helpful	Share less	Activity level	Cautiousness
Less helpful	Share less	Anxiety	Self-consciousness

Table 3. Examples of recommendations provided to the participants to simulate a recommender system.

Facet	User rating	Teammate rating	Recommendation 1
Dependability	Low	High	This teammate tends to be very responsible and highly accountable for their work. Make an effort to give notice if you're unable to deliver obligations on time.
Assertiveness	Mod.	Low	This teammate might prefer to hold back their opinions or stay in the background. Call on this teammate to ensure their viewpoint is heard so that they contribute to the team's success.
Self-efficacy	High	Low	This teammate tends to doubt their ability to get the job done. Make an effort to encourage this teammate to share skills they are confident in, and areas in which they need help.
Trust	Low	Mod.	This teammate is more trusting of others than you. Make sure not to take advantage of this trust by following through on expectations and remaining accountable.

Table 4. Study 4 examples of algorithmic and benefit recommendations for "self-efficacy."

Content	Occurrence
Algorithmic	Rationale for disclosure: Based on (1) how similar/different you and your teammate rank on this attribute and (2) similar users to your teammate have indicated they find receiving this information helpful. Rationale for recommendation: Based on (1) how similar/different you and your teammate rank on this attribute and (2) similar users to you have indicated they find receiving this information helpful.
Benefit	Rationale for disclosure: It is helpful for your teammates to understand how confident members are in their abilities which might influence roles and responsibilities. Rationale for recommendation: It is helpful for you to understand how confident members are in their abilities which might influence roles and responsibilities.

In addition to these measures, participants took a tailorable version of the Big Five personality assessment (Johnson, 2014). The survey asked the participants questions pertaining to the eight sub-scales that related to the experimental task (i.e., dependability, assertiveness, self-efficacy, trust, activity level, cautiousness, anxiety, and self-consciousness). This resulted in a personality questionnaire that contained 32 of the normal 120 questions.

3.4.2. Disclosure selection

Participants all received a “disclosure score” from 0 to 8. This disclosure score was a combined score of their willingness to share each of the eight personality results they received from the recommender system. Participants made a binary selection (i.e., yes or no) regarding whether or not they were willing to share such information. The disclosure score was the summation of these binary answers and was then used as a dependent variable.

3.4.3. System perception measures

After and while using the system, participants took six surveys related to their perceptions of the system. These measures include perceived accuracy of their personality results, perceived helpfulness of recommendations, privacy concern (Knijnenburg & Kobsa, 2014), trust competence (Benbasat & Wang, 2005; Komiak, 2003; Wang & Benbasat, 2007), and system satisfaction (Knijnenburg & Kobsa, 2013) using a 5-item Likert scale. Perceived accuracy of personality results was measured using a 1-question survey (“Regarding your personality results that were just displayed to you, how accurate do you think your results were?”) and was asked after personal personality results were displayed. Perceived helpfulness of recommendations was measured using a 1-question survey (“How helpful would it be to receive this recommendation?”) that was asked four times, once after each recommendation. The other four scales are established and validated surveys from previous studies, as cited above.

4. Results

4.1. Disclosure

To investigate the effect of explanation content on disclosure during system elicitation of personal information, conditions were simplified to those that provided algorithmic explanations during disclosure (Conditions 1 and 2), those that provided benefit explanations during disclosure (Conditions 5 and 6), and those that did not provide any explanations during disclosure (Conditions 3, 4, 7, and 8). Running an ANOVA, results indicate that there was a near-significant effect of providing benefit explanations during disclosure on disclosure, $b = 0.67$, $t(147) = 1.62$, $p_{one-tailed} = 0.054$. Meanwhile providing algorithmic explanations during disclosure had a positive effect on disclosure, $b = 0.21$, $t(147) = 0.51$, $p_{one-tailed} = 0.306$, though this was not significant. See Figure 2.

Thus, there are evident trends that providing explanations during disclosure increases disclosure, and providing benefit-related explanations during disclosure will increase disclosure. However, these relationships are only near-significant and not-significant.

4.2. Structural equation modeling

Structural Equation Modeling (SEM) was conducted to determine how various manipulations and factors relate to various perceptions of the system (Figure 3). SEM is a statistical analysis technique that measures and analyzes the quantitative relationships between observed and latent variables (Hoyle, 1995). SEM was selected due to the multitude of relationships that exist between the manipulations and dependent variables of this study (Biddle & Marlin, 1987). The inclusion of SEM addresses measurement error and increases the reliability of this analysis in assessing participant perceptions (Marks et al., 2005).

Before conducting the analysis, we conducted confirmatory factor analysis (CFA) and examined the validity and

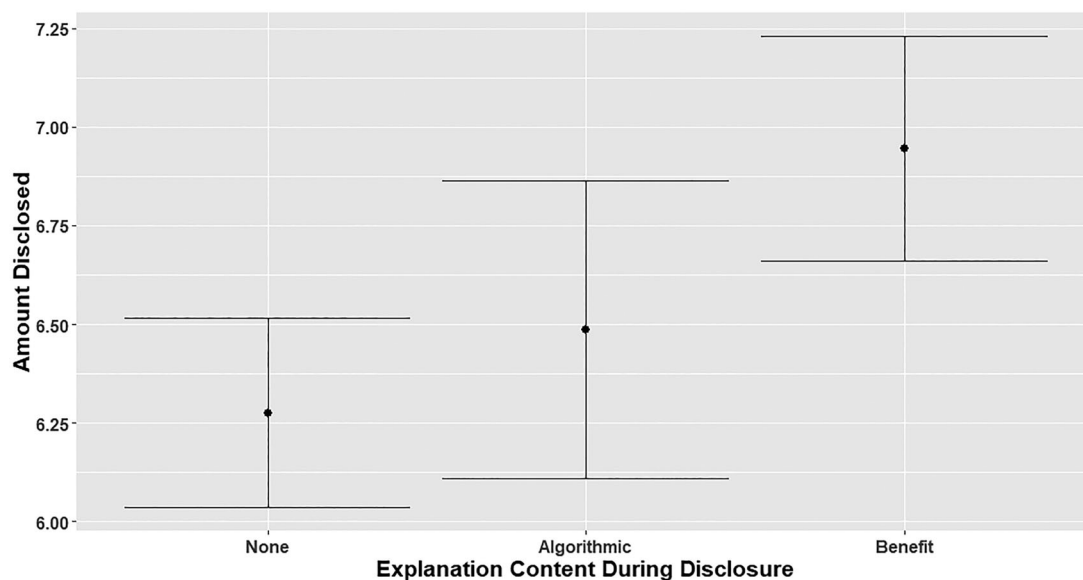


Figure 2. Disclosure by explanation content: Y-axis ranges from 0 to 8 items disclosed.

Table 5. Survey items per measurement with item factor loadings.

Measurement	Items	Factor loading
System satisfaction	The system has no real benefit to me.	0.726
	Using the system is annoying.	−0.841
	The system is useful.	0.810
	Using the system is a pleasant experience.	0.882
	Using the system makes me happy.	0.890
	Overall, I am satisfied with the system.	0.957
	I would recommend the system to others.	0.922
	I would use this system if it were available.	0.922
	I would pay \$2 to use this system.	–
	I would quickly abandon using this system.	−0.848
Privacy concern	It would take a lot of convincing for me to use this system.	−0.850
	I'm afraid the system discloses private information about me.	0.921
	The system invades my privacy.	0.966
	I feel confident that the system respects my privacy.	−0.906
Trust competence	This system is like a real expert in assessing teammates.	0.917
	This system has the expertise to understand my needs for understanding my teammates.	0.946
	This system has the ability to understand my needs for understanding my teammates.	0.871
	This system has good knowledge about my teammates.	0.834
Trust propensity	This system considers my needs and all important attributes of teamwork.	0.836
	It is easy for me to trust a person/thing.	0.949
	My tendency to trust a person/thing is high.	0.954
	I tend to trust a person/thing, even though I have little knowledge of it.	0.920
	Trusting someone or something is difficult for me.	−0.906

Two items from the system satisfaction measure were removed due to low loading (shown highlighted in gray without a factor loading in the table).

Table 6. A summary of correlations between every two factors.

	AVE	Satisfaction	Privacy concern	Trust competence	Trust propensity
System satisfaction	0.75	0.87	–	–	–
Privacy concern	0.87	−0.65	0.93	–	–
Trust competence	0.78	0.79	−0.45	0.88	–
Trust propensity	0.89	0.29	−0.32	0.27	0.94

The diagonal values represent the square root of the factor's average (e.g., the square root of system satisfaction's average is 0.87).

Bolded values represent the statistical diagonal, the square root of the factor's average, as defined in the table caption.

reliability scores of the constructs measured in this study with multiple items. Upon inspection of the CFA model, we removed two system satisfaction items including S1 (communality: 0.399) and S9 (communality: 0.372). The remaining items shared at least 44% of their variance with their designated construct. The final factor solution has a good fit ($\chi^2(183) = 382.661$, CFI = 0.921, TLI = 0.909, RMSEA: 0.085, 90% CI: [0.074, 0.097]). Factor loadings are shown in Table 5 with correlations between the factors listed in Table 6. To ensure the convergent validity of constructs, we examined the average variance extracted (AVE) of each construct. The AVEs were all higher than the recommended value of 0.50, indicating adequate convergent validity. To ensure discriminant validity, we ascertained that the square root of the AVE for each construct was higher than the correlations of the construct with other constructs.

The factors were then subjected to a structural model that also included the single-item measures and the experimental manipulations. Figure 3 presents the final trimmed model. The model's fit indices suggest a good fit: ($\chi^2(282) = 569.335$, CFI = 0.914, TLI = 0.903, RMSEA: 0.082, 90% CI: [0.073, 0.092]).

The model shows that providing benefit-related explanations has a positive effect on the perceived trust competence of the system ($\beta = 0.470$, $p_{\text{one-tailed}} = .030$) and a negative effect on system-specific privacy concerns ($\beta = -0.323$, $p_{\text{one-tailed}} = .028$) (see Figure 4). There was an interaction effect between benefit content explanations and the

occurrence of explanations during recommendations on trust competence ($\beta = -0.602$, $p = .080$). This means that compared to benefit-related explanations, algorithmic explanations had a negative effect on trust competence when provided during disclosure (see Figure 5). There was also an interaction effect between benefit content explanations and the occurrence of explanations during recommendations on perceived helpfulness ($\beta = 1.420$, $p = .085$). This means that the influence on perceived helpfulness increased for benefit-related explanations when they were provided during recommendations more than they increased for algorithmic-related explanations (see Figure 6).

Other factors related to system perceptions include an individual's trust propensity, which positively affects perceived trust competence ($\beta = 0.229$, $p = .013$) and a negative effect on privacy concern ($\beta = -0.273$, $p = .002$). A user's perception of the accuracy of their personality results has a positive effect on both their perceived trust competence in the system ($\beta = 0.211$, $p = .045$) and their perceived helpfulness of the system ($\beta = 0.682$, $p = .007$). Privacy concern has a negative effect on trust competence ($\beta = -0.292$, $p = .001$), which in turn has a positive effect on perceived helpfulness ($\beta = 1.203$, $p < .001$). Last, perceived helpfulness has a positive effect on user satisfaction with the system ($\beta = 0.067$, $p = .058$) as does trust competence ($\beta = 1.028$, $p < .001$), with privacy concern being related to reduced system satisfaction ($\beta = -0.535$, $p < .001$).

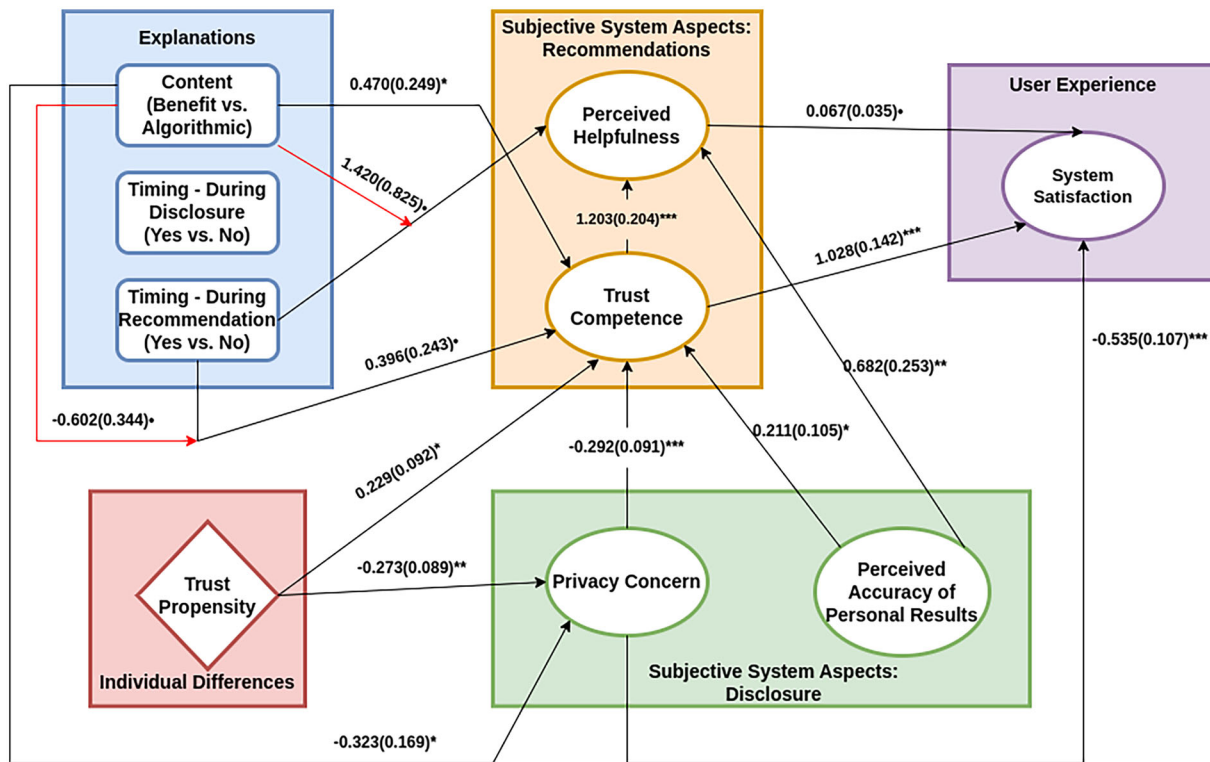


Figure 3. Structural model with near-significant and significant results ($p < .1$, * $p < .05$, ** $p < .01$, *** $p < .001$). numbers on the arrows represent the β coefficients with standard errors in parenthesis. black arrows indicate interaction effects.

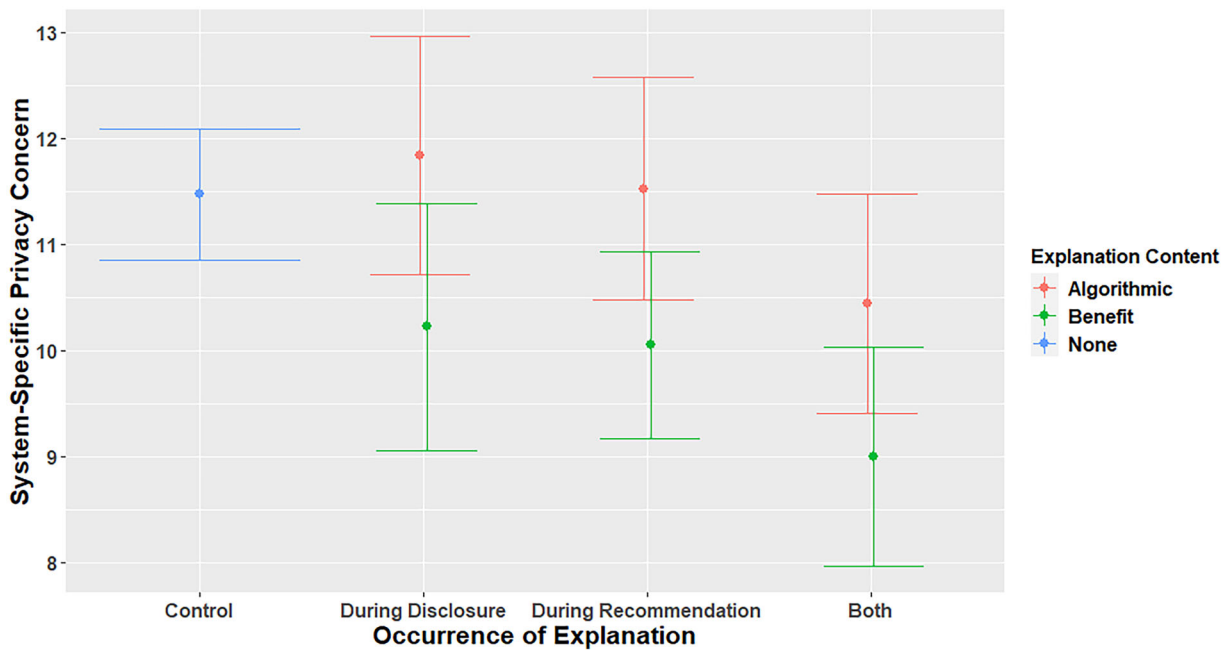


Figure 4. System-Specific Privacy concern by explanation occurrence and content: Y-axis ranges from 5 to 21.

5. Discussion

This study considers how information sharing recommendation systems can be used to support temporary teams, specifically in regard to the content and timing of explanations provided by the system. In this section, we will first discuss some of the key implications of those results, followed by design recommendations for such systems. The trends in our results reveal some very interesting points to consider in

designing a system that can both foster self-disclosure of personality types and acceptance of the recommendations provided. Of note is that the content of the explanations seemed to have a bigger impact on participant perceptions than timing. Such results mirror privacy research on users of social media, who pay special attention to the benefits they receive from a system inherently designed to share their information (Himeur et al., 2022). This also seems to support a key tenant of teamwork research, which is that one

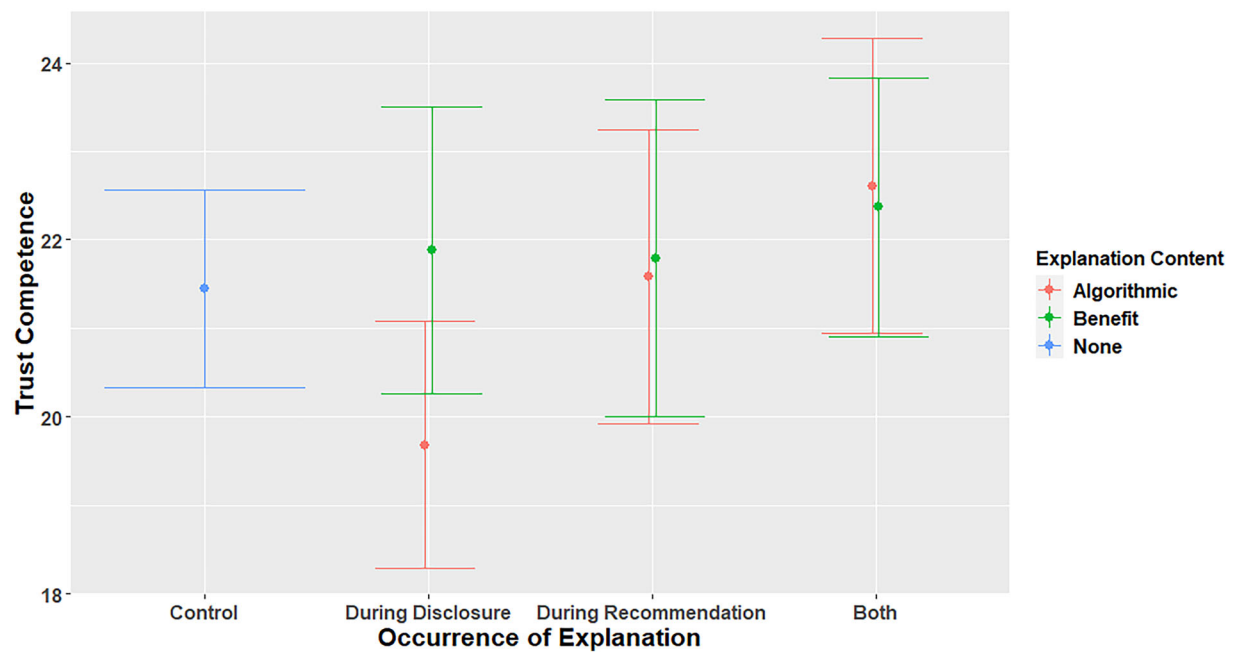


Figure 5. Perceived trust competence by explanation occurrence and content: Y-axis ranges from 5 to 35.

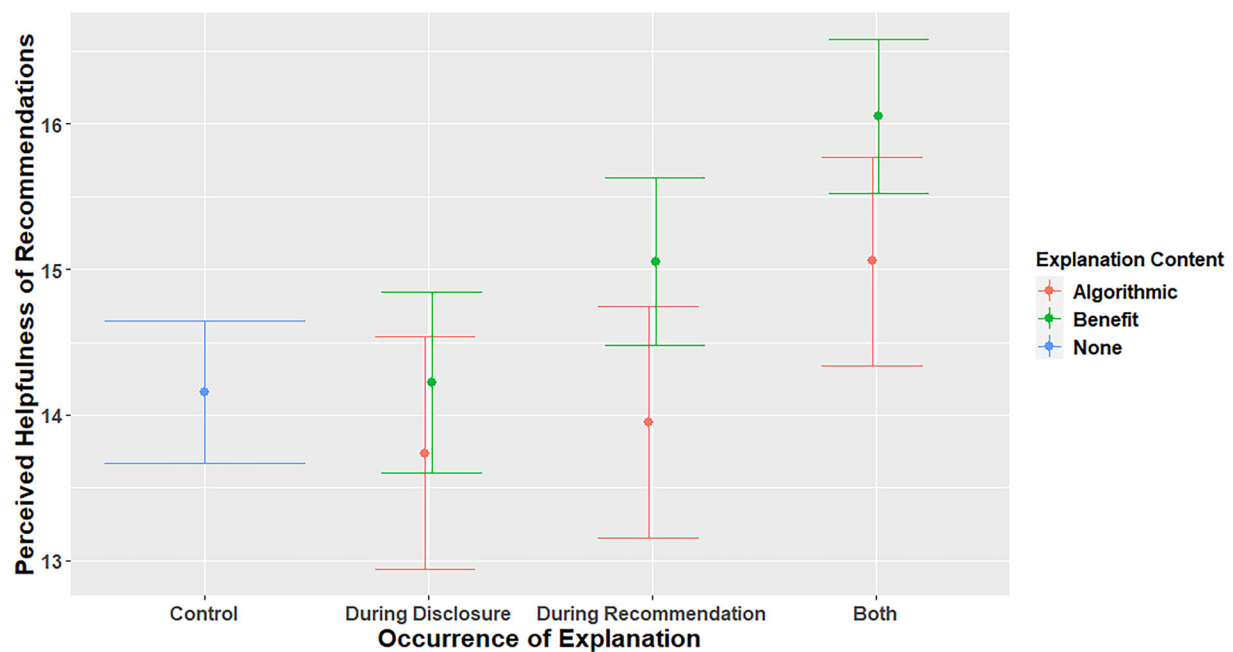


Figure 6. Perceived helpfulness by explanation occurrence and content: Y-axis ranges from 4 to 20.

must perceive the benefit of team membership before they commit to and put in work for the team (Ezzamel & Willmott, 1998). Our results indicate that this extends to accepting recommendations for how to interact with temporary team members.

Another very interesting trend in our data, which will be discussed in more detail in our first recommendation, is the importance of the system providing team members with personality assessments that they perceive as accurate. This makes sense, as on temporary teams, the information provided to the team members about each other is the basis on which they have to interact, due to a lack of time to build

relationships and a reputation. If a user perceives their own results to be incorrect, they are unlikely to trust what the system says about another person on the team to be accurate. This would be particularly true for student users on a class team, as these teams are often composed not only of individuals with different strengths and weaknesses but also with different levels of commitment (Colbeck et al., 2000). The prospective usefulness of a recommender system that could aid these very short-term teams in quickly understanding what and how much each member will bring to the team frames it as a good use case for exploring the following two recommendations.

5.1. Fostering trust in the validity of the information before using the system

The perception that the personality information was accurate was an important factor for both trust and perceived helpfulness. This trust needs to be addressed/promoted before using the system, as information sharing recommender system research shows the perceived accuracy of the information shablack significantly affects user perceptions of the system (Wu et al., 2022). In the presented SEM model, *perceived accuracy of personality results* had a significant effect on both perceived trust competence and helpfulness of the system. These findings echo the findings of previous research that has highlighted how some users of personality-sharing systems perceive the Big Five personality results to be inaccurate, which undermines their perceptions of the system (Musick et al., 2023). As such, there is a mismatch of perceptions and prior research, as prior research has emphasized the accuracy and stability of the measure on temporary teams (Stidham, Flynn, et al., 2018; Stidham, Summers, et al., 2018). This is important to consider for this study, because in these types of teams, team members do not have time to naturally gain an understanding of one another, and so they need to rely more on the information they receive about the other members than on first-hand interactions and observations.

As using an information sharing recommender system will likely be a novel experience to many users at this point in time, there will likely be a certain amount of uncertainty regarding perceptions of trust and helpfulness of such a system, to begin with. Therefore, there is an opportunity to create an intervention to increase perceptions of the accuracy of personality results so as to avoid creating additional uncertainty or concerns about trust. We recommend the use of training, whether digital or personal, to describe the source of personality traits that the system is using, as well as the accuracy and validity of the data source before using the system. As temporary teams have little time to conduct on-the-job training, this should occur as part of a team member's onboarding process, or upon the formation of a brand-new team. Many organizations have a formalized process for training members on the systems they need to use to complete their role where this training would easily fit, such as during the certification process computer security specialists undergo when they start a new job (Kouki et al., 2019). As providing evidence as to the accuracy of data sources for AI can increase trust for users (Arnold et al., 2019), this recommendation should result in both increased perceived trust and helpfulness of the system and therefore increased system satisfaction. For the case of students combining as a team to complete a course project, it is likely few have used a recommender system before, and that heightens the importance of training them on the system. When students receive their assignment, they could also be assigned homework that trains them on the recommender system and the personality tests it uses before each member actually using it and beginning the project.

5.2. Creating an adaptive and intelligent explanation system

An interesting trend, though not significant in the model, was the interaction effect between *occurrence—during disclosure* and *occurrence—during recommendations*. Specifically, there was a trend to improve system perceptions when explanations were provided during both disclosure and recommendations. This can be seen in Figure 6 as the conditions that provided explanations at both times resulted in the highest perceived helpfulness (for both algorithmic and benefit explanations) and can be seen in Figure 4 as the conditions that provided benefit explanations at both times resulted in the lowest perceived privacy concerns.

Based on this trend, a reactionary design recommendation would be to provide both benefit and algorithmic recommendations at both times for every disclosure and every recommendation for an information sharing recommender system; however, prior research has emphasized the importance of providing the right content and level of detail depending on the goals of the explanation and the type of user (Mohseni et al., 2021). This is likely due to designers needing to consider the cognitive overload or confusion possible when providing too much or not the right type of information (Gedikli et al., 2014; Kizilcec, 2016; Zhao et al., 2019). Considering our use case, a student who is part of three different temporary teams for classes may have less time and investment in considering the explanations than a new employee whose paycheck is dependent on their successful integration with the team. With this in mind, recent research has suggested making personalized explanations and explanations with different levels of detail depending on the user (Chatti et al., 2022).

Therefore, we suggest for this context creating an adaptive and intelligent user interface to provide explanations. In this design, both benefit and algorithmic explanations will always be available to users if they interact with the user interface to request an explanation. This mirrors recent research on the importance of personalizing both the levels of disclosure to a system (Chen et al., 2023) and the levels of explanations to best suit the user's needs (Chen et al., 2019). As such, the system should consider user-tailorblock privacy settings and be adapted to provide explanations when items are more important to disclose or adapt to user profiles to proactively give benefit-related explanations to users who are more likely to have privacy concerns (Knijnenburg et al., 2021). This would be an inblackibly important feature for organizations and areas in which individuals routinely become a part of temporary teams, such as a university system. While their time on a specific team may be short, their profile can still build and adapt based upon multiple iterations of participation from class to class. In such a system, the recommender system will give the right explanation at the right time for the appropriate user.

5.3. Limitations and future work

This study serves as a useful foundational study for investigating occurrence and content for explanations in this

context. However, a few notable limitations provide an opportunity for future research. First, this study used vignette representations of a recommender system and hypothetical teams, and future work should extend the results by examining actual recommender systems and real-world teams. Notably, using actual teams would likely increase the realism of disclosure behaviors and allow for the consideration of historical disclosure behaviors over time. Additionally, this work only explored explanation content through the inclusion of benefit-focused explanations. Given the significant results of this work, future efforts should be inspired to explore additional types of content that should exist within explanations. Finally, all participants self-reported answers to the individual differences measures used in this study to include their disclosure score. Research that includes unbiased and historically based data on the participants' personalities and disclosure tendencies would enhance the validity of the findings. This would require a longitudinal study over multiple temporary teams, which makes our proposed use case of university courses that use team projects also an appropriate next step for this line of research.

6. Conclusion

Previous HCI work has focused on how temporary teams can be supported during team formation (Harris et al., 2019). A challenge for these teams is that they do not have the benefit of developing team norms, TMMs, and understanding of each other's strengths and weaknesses. Information sharing recommender systems can be used to help these teams work together and excel (Yang et al., 2015), but those systems are dependent on the willingness of the team members to truthfully disclose information to the system (Musick et al., 2023). In this study, we showed how explanations can be used during both the disclosure and the recommendation process to encourage users to disclose more information and increase trust in the system, as well as how benefit-related explanations tend to decrease team members' privacy concerns. The results of this study provide important design implications for researchers and practitioners who wish to develop information sharing recommender systems that promote trust in and disclosure of the system while minimizing user privacy concerns.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendix A: Personality surveys

A.1. Modified Big Five personality assessment

Sources: Goldberg (1999) and Johnson (2014)

Please use the rating scale below to describe how accurately each statement describes you (5-point Likert, Very Accurate $\Longleftrightarrow$ Very Inaccurate).

Note: the questions have been organized by facet for this appendix but were not grouped as such when presented to users.

(1) Extraversion

Activity level

Am always busy.
Am always on the go.
Do a lot in my spare time.
Like to take it easy.*

Assertiveness

Take charge.
Try to lead others.
Take control of things.
Wait for others to lead the way.*

Cheerfulness

Radiate joy.
Have a lot of fun.
Love life.
Look at the bright side of life.

Excitement-Seeking

Love excitement.
Seek adventure.
Enjoy being reckless.
Act wild and crazy.

Friendliness

Make friends easily.
Feel comfortable around people.
Avoid contacts with others.*
Keep others at a distance.*

Gregariousness

Love large parties.
Talk to a lot of different people at parties.
Prefer to be alone.*
Avoid crowds.*

(2) Emotionality

Anxiety

Worry about things.
Fear for the worst.
Am afraid of many things.
Get stressed out easily.

Frustration

Get angry easily.
Get irritated easily.
Lose my temper.
Am not easily annoyed.*

Immoderation

Go on binges.
Rarely overindulge.*
Easily resist temptations.*
Am able to control my cravings.*

Melancholic

Often feel blue.
Dislike myself.
Am often down in the dumps.
Feel comfortable with myself.*

Self-Consciousness

Find it difficult to approach others.
Am afraid to draw attention to myself.
Only feel comfortable with friends.
Am not bothered by difficult social situations.*

Vulnerability

Panic easily.
Become overwhelmed by events.
Feel that I'm unable to deal with things.
Remain calm under pressure.*

(3) Conscientiousness**Achievement-Striving**

Work hard.
Do more than what's expected of me.
Do just enough work to get by.
Put little time and effort into my work.

Cautiousness

Jump into things without thinking.
Make rash decisions.
Rush into things.
Act without thinking.

Dependability

Keep my promises.
Tell the truth.
Break rules.
Break my promises.

Orderliness

Like to tidy up.
Often forget to put things back in their proper place.
Leave a mess in my room.
Leave my belongings around.

Self-Efficacy

Complete tasks successfully.
Excel in what I do.
Handle tasks smoothly.
Know how to get things done.

Self-Discipline

Am always preoccupied.
Carry out my plans.
Waste my time.
Have difficulty starting tasks.

(4) Agreeableness**Altruism**

Love to help others.
Am concerned about others.
Am indifferent to the feelings of others.
Take no time for others.

Cooperation

Love a good fight.

Yell at people.
Insult people.
Get back at others.

Modesty

Believe that I am better than others.*
Think highly of myself.*
Have a high opinion of myself.*
Boast about my virtues.*

Morality

Use others for my own ends.
Cheat to get ahead.
Take advantage of others.
Obstruct others' plans.

Sympathy

Sympathize with the homeless.
Feel sympathy for those who are worse off than myself.
Am not interested in other people's problems.
Try not to think about the needy.

Trust

Trust others.
Believe that others have good intentions.
Trust what people say.
Distrust people.

(5) Openness**Adventurousness**

Prefer variety to routine.
Prefer to stick with things that I know.
Dislike changes.
Am attached to conventional ways.

Artistic interests

Believe in the importance of art.
See beauty in things that others might not notice.
Do not like poetry.
Do not enjoy going to art museums.

Imagination

Have a vivid imagination.
Enjoy wild flights of fantasy.
Love to daydream.
Like to get lost in thought.

Intellect

Love to read challenging material.
Avoid philosophical discussions.
Have difficulty understanding abstract ideas.
Am not interested in theoretical discussions.

Liberalism

Believe that criminals should receive help rather than punishment.
Believe that there is no absolute right or wrong.
Believe too much tax money goes to support artists.
Believe that we should be tough on crime.

Sentimentality

Experience my emotions intensely.
Feel others' emotions.
Rarely notice my emotional reactions.
Don't understand people who get emotional.

A.2. Conflict management styles

Sources: Rahim (1983) and Thomas (2008)
Please use the rating scale below to describe how accurately each statement describes you (5-point Likert scale, Very Accurate $\Longleftrightarrow$ Very Inaccurate).
Note: the questions have been organized by conflict management style for this appendix but were not grouped as such when presented to users.

Integrating

I try to investigate an issue with others to find a solution acceptable to everyone.

I try to integrate my ideas with others to come up with a decision jointly.
 I try to work with others to find solutions to a problem which satisfies everyone's expectations.

I exchange accurate information with others to solve a problem together.
 I try to bring everyone's concerns out in the open so that the issue can be resolved in the best possible way.

I collaborate with others to come up with decisions acceptable to everyone.
 I try to work with others for a proper understanding of a problem.

Accommodating

I generally try to satisfy the needs of everyone.

I usually accommodate the wishes of others.

I give in to the wishes of others.

I usually allow concessions to others.

I often go along with the suggestions of others.

I try to satisfy the expectations of everyone.

Compromising

I try to find a middle course to resolve an impasse.

I usually propose a middle ground for breaking deadlocks.

I negotiate with my supervisor so that a compromise can be reached.

I use "give and take" so that a compromise can be made.

Avoiding

I attempt to avoid being "put on the spot" and try to keep my conflict with others to myself.

I usually avoid open discussion of my differences with others.

I try to stay away from disagreements with others.

I avoid encounters with others.

I try to keep my disagreements with others to myself to avoid hard feelings.

I try to avoid unpleasant exchanges with others.

Dominating

I use my influence to get my ideas accepted.

I use my authority to make a decision in my favor.

I use my expertise to make a decision in my favor.

I am generally firm in pursuing my side of the issue.

I sometimes use my power to win a competitive situation.

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